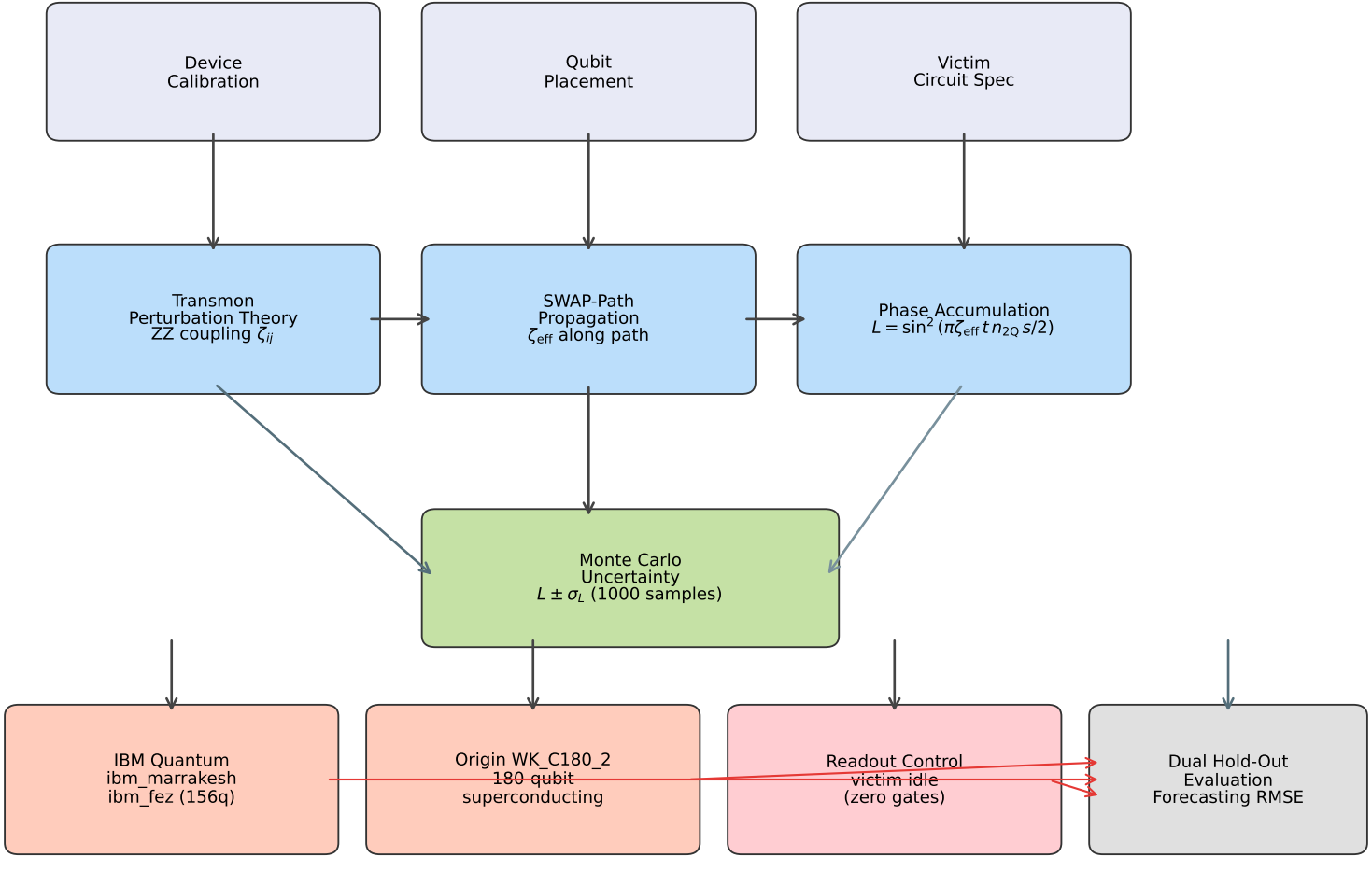CrossEcho Pipeline Architecture